A070 (provisional) — Absolute Time, Swirl Clocks and Operational Distance

Executed analytical / numerical falsifier pilot v0.1.0

SST Workbench — exploratory science instance

18 October 2026
DEVELOPMENT / NOT CERTIFIED / NOT PHYSICALLY BLIND

Status and limitations. This is a self-contained model-diagnostic pilot, not a frozen SST Framework v1.0.7 instance. Source-native finite-core PKLSA input, experimentally determined photon propagation, a physical rod model and astronomical likelihood tests are not available here. Synthetic-controls cannot certify gravity or cosmology. The catalog ID A070 is unverified against local Workbench registries.

1 The fundamental distinction: absolute and operational quantities

The SST hypothesis is a Euclidean, incompressible fluid evolving in one absolute time parameter t:

$$\nabla \cdot \mathbf{u} = 0, \qquad \frac{D\mathbf{u}}{Dt} = -\frac{\nabla p}{\rho_f}. \tag{1}$$

Clock i has internal vortex phase θ_i and inferred local Swirl-Clock time:

$$\frac{d\theta_i}{dt} = \Omega_i(t), \qquad \frac{d\tau_i}{dt} = f_i(t) = \frac{\Omega_i(t)}{\Omega_{i,0}}, \qquad \tau_i(t) = \int_0^t f_i(s)\, ds. \tag{2}$$

Euclidean separation D_{abs}, two-way radar separation D_{rad}, and rod-count separation D_{rod} are *distinct physical constructions*. Inferring the last two requires independent signal/rod physics. Absolute time is not a universal synchrony requirement for atomic clocks.

2 An exact stretching Euler control

Use the nonuniform affine velocity field

$$\mathbf{u} = \begin{pmatrix} -sx/2 - \Omega(t)y \\ \Omega(t)x - sy/2 \\ sz \end{pmatrix}, \qquad \Omega(t) = \Omega_0 e^{st}, \quad s > 0. \tag{3}$$

It obeys $\nabla \cdot \mathbf{u} = 0$ and $\boldsymbol{\omega} = (0, 0, 2\Omega)$. Direct substitution into the Euler equation yields the exact pressure

$$p = p_0 - \frac{\rho_f}{2}\left[\left(\frac{s^2}{4} - \Omega(t)^2\right)(x^2 + y^2) + s^2 z^2\right]. \tag{4}$$

The moving core is stretched along z and compressed transversely, preserving local material volume:

$$L_z/L_{z0} = e^{st}, \quad a/a_0 = e^{-st/2}, \quad \Omega/\Omega_0 = e^{st}. \tag{5}$$

The analytic and Python FP64 acceleration matrices match in the reported benchmark. **Critical caveat:** Eq. (3) is not a finite-energy global vortex network or a dynamically certified knotted filament. Its infinite-domain behavior makes it suitable only as an analytic control. For numerical illustration only, $\Omega_0 = |\mathbf{v}_{\circlearrowleft}|/r_c = 7.76344 \times 10^{20}\,\mathrm{s}^{-1}$, $s = 0.12/\mathrm{Gyr} = 3.80257 \times 10^{-18}\,\mathrm{s}^{-1}$ and $\rho_f = 7 \times 10^{-7}\,\mathrm{kg\,m}^{-3}$. The clock frequency is a hypothesis about core rotation, not a resolved PKLSA mode.

3 Gate results: absolute distance versus clock and rod measures

G1: Clock-only acceleration. For fixed $D_{\rm abs} = D_0$, under a constant absolute-coordinate signal speed c and *short-baseline/quasi-static* two-way ranging:

$$\tau = \frac{e^{st} - 1}{s}, \qquad D_{\rm rad} = f(t) D_0 = D_0(1 + s\tau), \qquad \boxed{\frac{d^2 D_{\rm rad}}{d\tau^2} = 0.} \tag{6}$$

Thus this particular physical exponential stretching law does **not** produce an accelerating operational distance from a fixed Euclidean separation. A derivative-on-grid test has maximum dimensionless second derivative 5.42×10^{-12} after rescaling distance by D_0 and local time by a Gyr. This result does not apply to cosmological light-travel distances, where full signal and clock history must be integrated.

G2: Rod–clock–signal closure. If a material rod is the core diameter $a(t)$, measured rod-count distance is $D_{\rm rod} = D_{\rm abs} e^{st/2}$, whereas radar gives $D_{\rm rad} = D_{\rm abs} e^{st}$. Therefore

$$\frac{c_{\rm measured}}{c} = \frac{1}{f(a/a_0)} = e^{-st/2}. \tag{7}$$

At one Gyr, with $s\,\mathrm{Gyr} = 0.12$, this gives 0.9417645: a 5.82% local measured speed drift under this *specific* trio of assumptions. A consistent SST photon law or different physical rod may change this; no unconditional falsification is claimed.

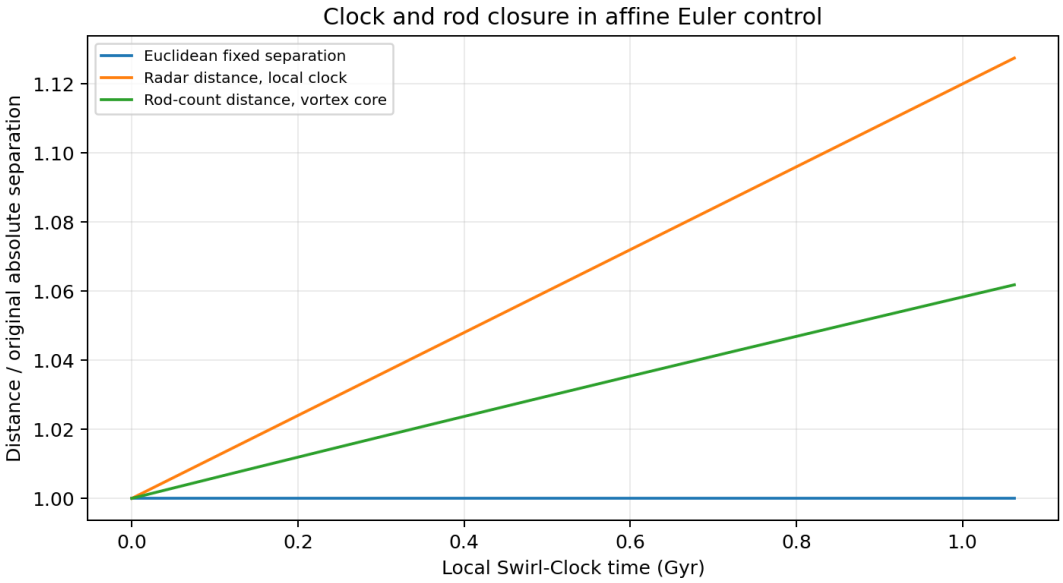


Figure 1: The same fixed Euclidean baseline can have different operational lengths under two independently proposed measurement recipes.

G3: Isotropy. A pair of material tracers separated along the z axis moves as $D_z = D_0 e^{st}$, while a transverse pair has $D_x = D_0 e^{-st/2}$. The volume ratio $D_z D_x^2 / D_0^3 = 1$. Radar estimates grow as e^{2st} in z but $e^{st/2}$ transversely, so a generically *isotropic* apparent expansion is not recovered. In local time, $D_{\mathrm{rad},z} = D_0(1+s\tau)^2$ and $D_{\mathrm{rad},x} = D_0(1+s\tau)^{1/2}$; these have opposite signs of second derivative.

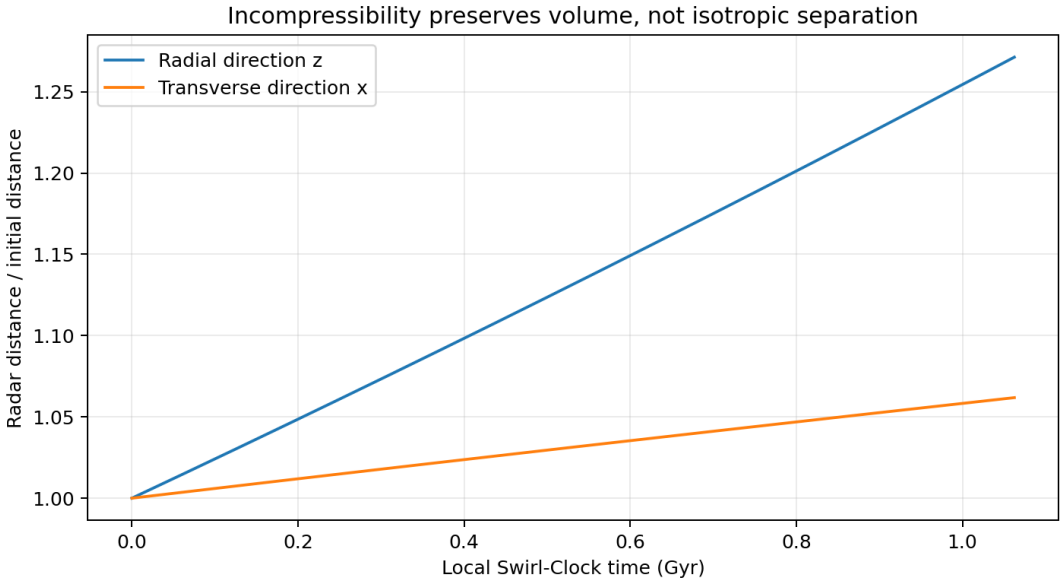


Figure 2: An exact incompressible affine flow generates axis-dependent separations: longitudinal expansion and transverse contraction.

4 Optical-distance hypothesis: synthetic EdS against synthetic flat ΛCDM

A static background does not have a derived SST photon/atomic-redshift law yet. Instead, as a *mathematical identifiability test*, compare the standard toy curves (not observations):

$$D_L(z) = (1+z)\frac{c}{H_0}\int_0^z \frac{dq}{E(q)}, \quad E_{\mathrm{EdS}} = (1+z)^{3/2}, \quad E_{\Lambda\mathrm{CDM}} = \sqrt{0.3(1+z)^3 + 0.7}. \tag{8}$$

For phenomenological optical depth $\mathcal{T}(z)$, $F_{\mathrm{obs}} = F_{\mathrm{EdS}} e^{-\mathcal{T}(z)}$ and $D_{L,\mathrm{obs}} = D_{L,\mathrm{EdS}} e^{\mathcal{T}/2}$. The exactly constructed target-dependent law is

$$\mathcal{T}_*(z) = 2\ln\left(\frac{D_{L,\Lambda\mathrm{CDM}}}{D_{L,\mathrm{EdS}}}\right). \tag{9}$$

It matches by construction and therefore has no predictive validity. At $z = 1$, $\mathcal{T}_*(1) = 0.550574$, corresponding to an extra 42.34% flux suppression compared to EdS and a distance-duality factor $\eta = e^{\mathcal{T}_*/2} = 1.31691$ (if D_A remains unchanged). Such large effects require independent observational testing.

G5: Low-z fit and independent high-z synthetic holdout. A restrictive one-parameter response $\mathcal{T}(z) = \alpha z$ was fit at $z \in \{0.1, 0.2, 0.35, 0.5\}$ and evaluated at $z \in \{0.8, 1.0, 1.5, 2.0\}$ without refitting. It gives $\alpha = 0.778159$ and held-out RMS magnitude residual 0.553317 mag, against exploratory preset acceptance 0.020 mag. This **fails the synthetic holdout**, not a real cosmic observation. EdS and ΛCDM already both have their own cosmological time dilation; opacity alone is not an SST origin of redshift or time stretching.

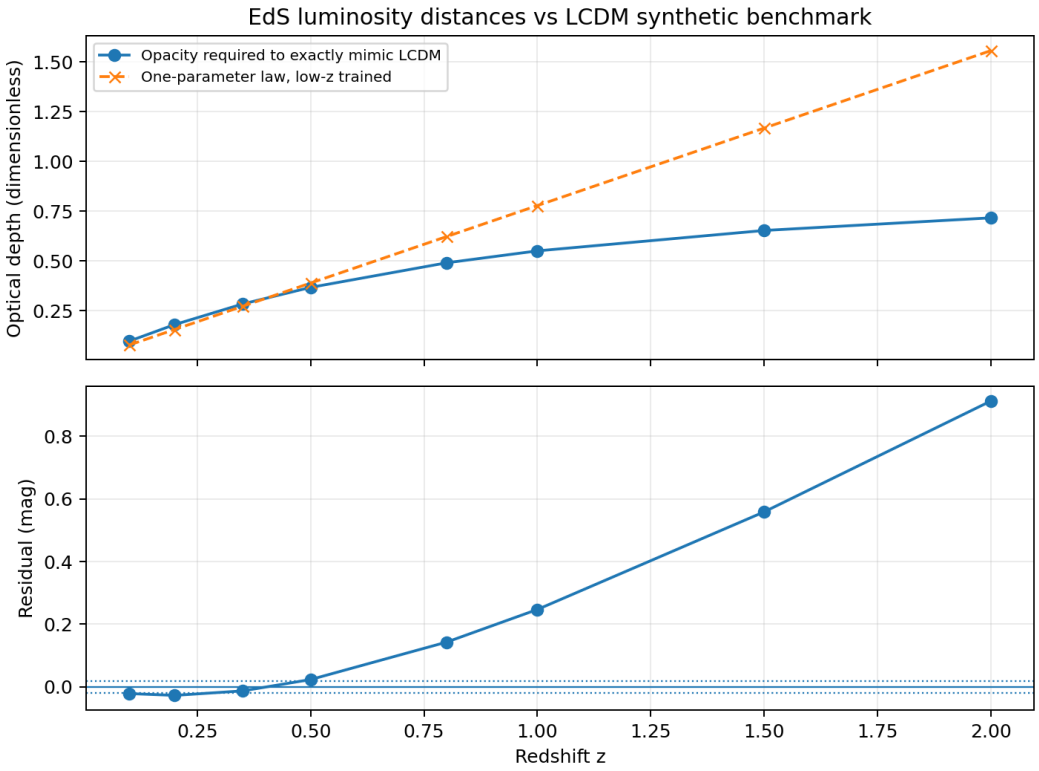


Figure 3: Exactly constructed target-dependent opacity (identity only) compared with the held-out extrapolation of one simple opacity law.

5 Audit-grade gate ledger

Gate	Numerical result	Scope / status
G0	PASS_REFERENCE	Exact solenoidal affine Euler control, not a finite-energy 3D knot.
G1	NULL_RETAINED	Clock-only positive acceleration fails for fixed absolute baseline in this model.
G2	FAIL_MODEL_CLOSURE	Clock + core radius + constant absolute signal speed imply 5.82% local signal-speed variation over chosen benchmark.
G3	FAIL_ISOTROPIC	Axial expansion and transverse contraction, material volume exactly preserved.
G4	PASS_IDENTITY_ONLY	Target-constructed opacity reproduces target by definition.
G5	FAIL_SYNTHETIC_HOLDOUT	0.553317 mag holdout RMS exceeds 0.020 mag synthetic threshold.
G6	UNRESOLVED	No resolved source-native finite-core PKLSA and no independent photon/rod physics or real observation likelihood.

6 Next physically discriminating test

Generate a frozen v1.0.7 instance only after audit of the Workbench registry ID. Supply source-native finite-core vortex seeds and independently propagated internal perturbation modes. Evolve them in common absolute t; determine $f_i(t)$ from modes, not a fitted relativistic factor; determine material length standards from resolved structure and light propagation from an independently specified field equation. Separate statistical evidence for distance–redshift, redshift–time dilation, BAO angular/radial distances and lensing. Never promote a synthetic-control PASS to a physical SST claim.

The framework v1.0.7 reference was verified in Drive on this date; its frozen generator, blind policies, CPU certification and PDF reporting were *not* executed as part of this standalone pilot. The code is source-ready for controlled integration.

References

References

[1] G. K. Batchelor (1967), *An Introduction to Fluid Dynamics*, Cambridge University Press. DOI: 10.1017/CBO9780511800955.

[2] I. M. H. Etherington (1933), "On the Definition of Distance in General Relativity," *Philosophical Magazine* **15**, 761–773. DOI: 10.1080/14786443309462220.

[3] A. G. Riess et al. (1998), "Observational Evidence from Supernovae for an Accelerating Universe and a Cosmological Constant," *Astronomical Journal* **116**, 1009–1038. DOI: 10.1086/300499.

[4] DESI Collaboration, *DESI DR2 cosmological results and documentation* (2025–2026), `https://data.desi.lbl.gov/doc/papers/`.

[5] H. Chaudhary, S. Capozziello, V. K. Sharma et al. (2026), "Evidence for evolving dark energy from DESI DR2 BAO and Pantheon$^+$, DES-Dovekie, and Union3," *European Physical Journal C* **86**, 564. DOI: 10.1140/epjc/s10052-026-15806-w.